AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 19 · STUDENT LESSON

Sampling and Representation

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Define the population. Audit selection. Report nonresponse.

Distinguish a calculated respondent proportion from evidence about everyone the question concerns.

Keep one habit: Question and frame → sample count → invitation/response denominators → independent plan.

MAIN LESSON · SECTION 1

Whose preferences can this survey describe?

> Separate the target population, invited sample and actual respondents.

Your learning goals

I can define a question and target **population**.

I can describe how a **sample** was selected.

I can limit conclusions to supported groups.

Recall an observed proportion

- Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

- If 6 of 10 **response**s choose Reading, what is the observed proportion?

Your end goal

By the end, I can define a **population**, selection frame and non**response** report for a survey.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Missing Data. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Over**fitting** Intuition. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Types of Classification Error. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

Week 19: Sampling and **Representation**

A **sample** must reflect the people and situations where the system will be used.
</details>



THINKER MISSION

Audit a playground survey collected from one class.

Thinker name and date:

> **Write or draw your reasoning and evidence.**

Readiness for the connected explanation

Readiness for Graphs and honest scales

Counts are 8 and 10. What is their absolute difference?

> **My readiness explanation**

This week's end goal

MAIN LESSON · SECTION 2

Define the question before choosing people to ask

> The coding club is one group, not the entire grade.

The target and convenient-sample problem

A fictional school wants to know which activity Grade 7 students prefer. Surveying only the coding club is easy, but the question concerns every Grade 7 student.

First define the **population**, selection **process**, question and **response** options. No real student data is required for this exercise.

▼ **Optional graph extension · A different eighteen-response poll**

A fictional poll records 8 votes for Drawing and 10 for Building. A **bar chart** should make lengths proportional to counts, starting at zero.

A chart starting at 7 makes the visible bars 1 and 3 units tall and can exaggerate the difference.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Notice → Predict → Explain

Begin with visible **evidence**. Do not explain more than the picture and information support.



MYSTERY

The learners must use sampling and **representation** to solve a problem. What information do they have? What decision or calculation do they make? What would count as **evidence** that it worked?

Three **observation**s (what you can point to):

> **Write or draw your reasoning and evidence.**

My **prediction** and the clue supporting it:

> **Write or draw your reasoning and evidence.**

A closer explanation

Apply the connected idea: Graphs and honest scales

MAIN LESSON · SECTION 3

Count club responses without turning them into a whole-grade estimate

> Code can calculate an observed count without fixing coverage.

Predict and run the three-response script

```
rows = [{"group": "club", "choice": "Code"},
        {"group": "club", "choice": "Read"},
        {"group": "club", "choice": "Code"}]
code_votes = 0
for row in rows:
    if row["choice"] == "Code":
```

```
        code_votes = code_votes + 1
print(code_votes, len(rows))
```

Match target, sample and observed result

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Target	All Grade 7 students	Population
Available data	Three club responses	Sample
Observed result	2 of 3 choose Code	Describes respondents

Read each row from left to right and connect the evidence to the stated rule.

Separate invitation from response

The result 2 of 3 describes these fictional club **response**s. It cannot by itself estimate the whole grade reliably. A better plan starts from a complete grade roster, uses a fair selection method and records non**response**.

Random selection gives each listed student a chance; it does not guarantee every small **sample** is representative.

The **population** is everyone the question concerns. The **sample** is the smaller group whose **response**s you actually collect.

If the question is about all Grade 7 students, a coding-club **sample** leaves other students out before the poll even starts. Separate the invitation **process** from the **response process**: some invited students may not reply.

A report needs both counts so readers can see whose preferences might be missing.

Explain the deciding limit

- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

▼ Optional graph extension · Complete text-chart script

```
counts = {"Drawing": 8, "Building": 10}
for label in counts:
    print(label, "#" * counts[label], counts[label])
```

Follow the evidence

STEP OR CASE	EVIDENCE	MEANING
Drawing	########	8 votes
Building	##########	10 votes

Read each row from left to right and connect the evidence to the stated rule.

The text chart uses one # per vote and a zero **baseline**. The second category has 2 more votes and 1.25 times as many votes, not three times as many.

Include the poll question, date or collection occasion and number of **response**s when presenting a real graph.

With a zero **baseline**, eight votes make a bar eight units high and ten votes make a bar ten units high. The difference is two votes.

If the drawing begins at seven, only the extra one and three units are visible, making the second bar look three times as tall. That picture describes the part above seven, not the full counts.

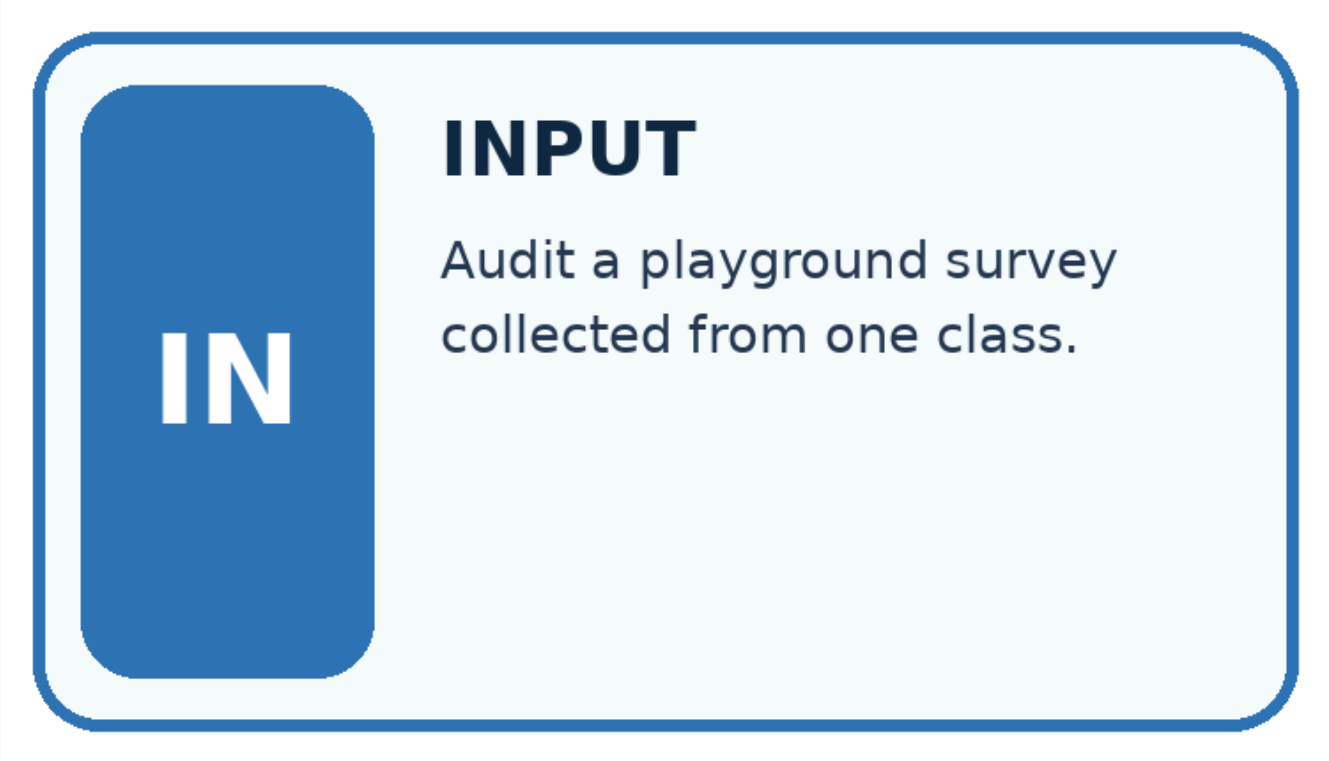
Use one **scale**, label the units and show the counts.

▼ **Earlier wording, alternatives and picture transcripts**

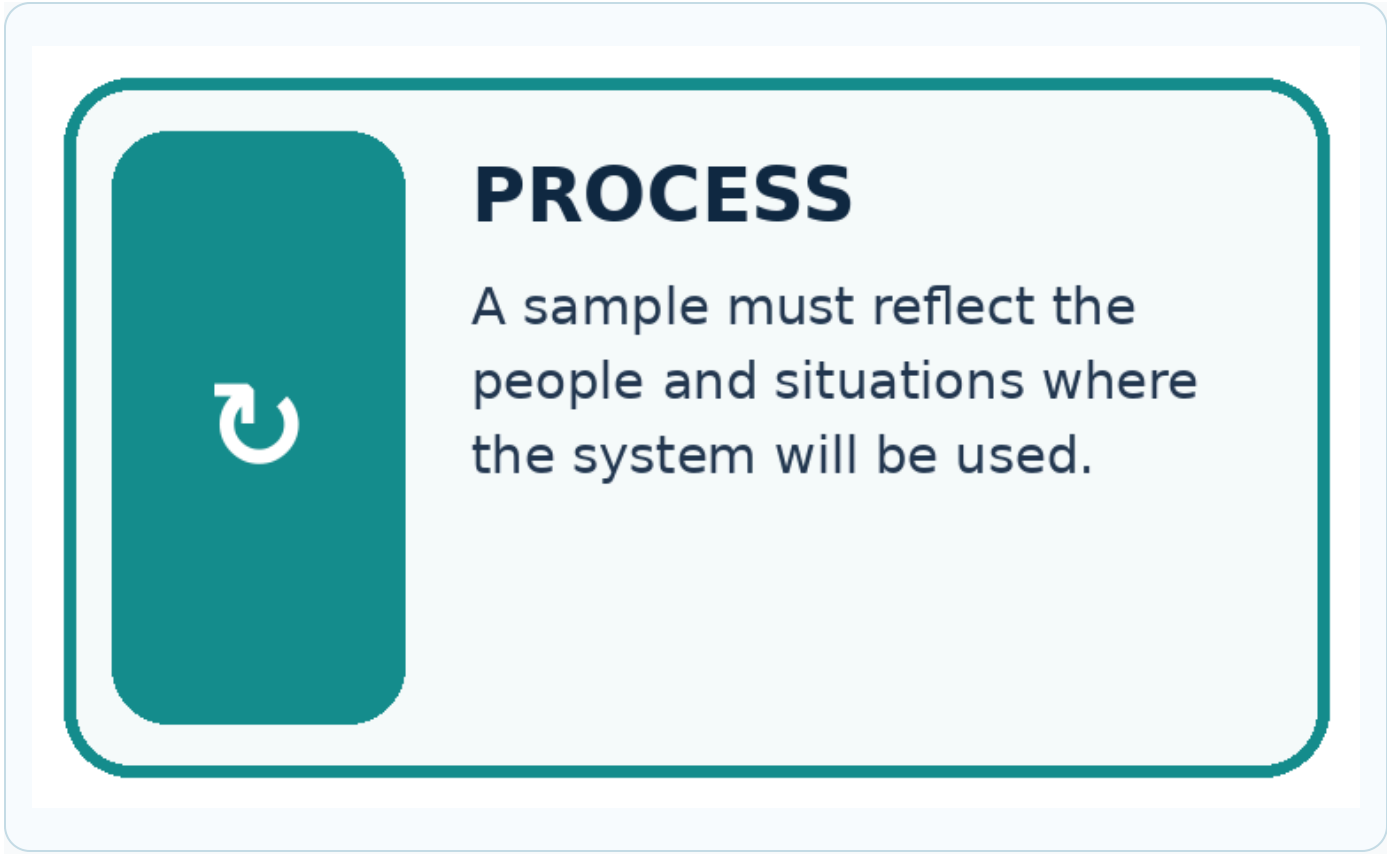
VISUAL **MECHANISM**

Input → **Process** → **Output**

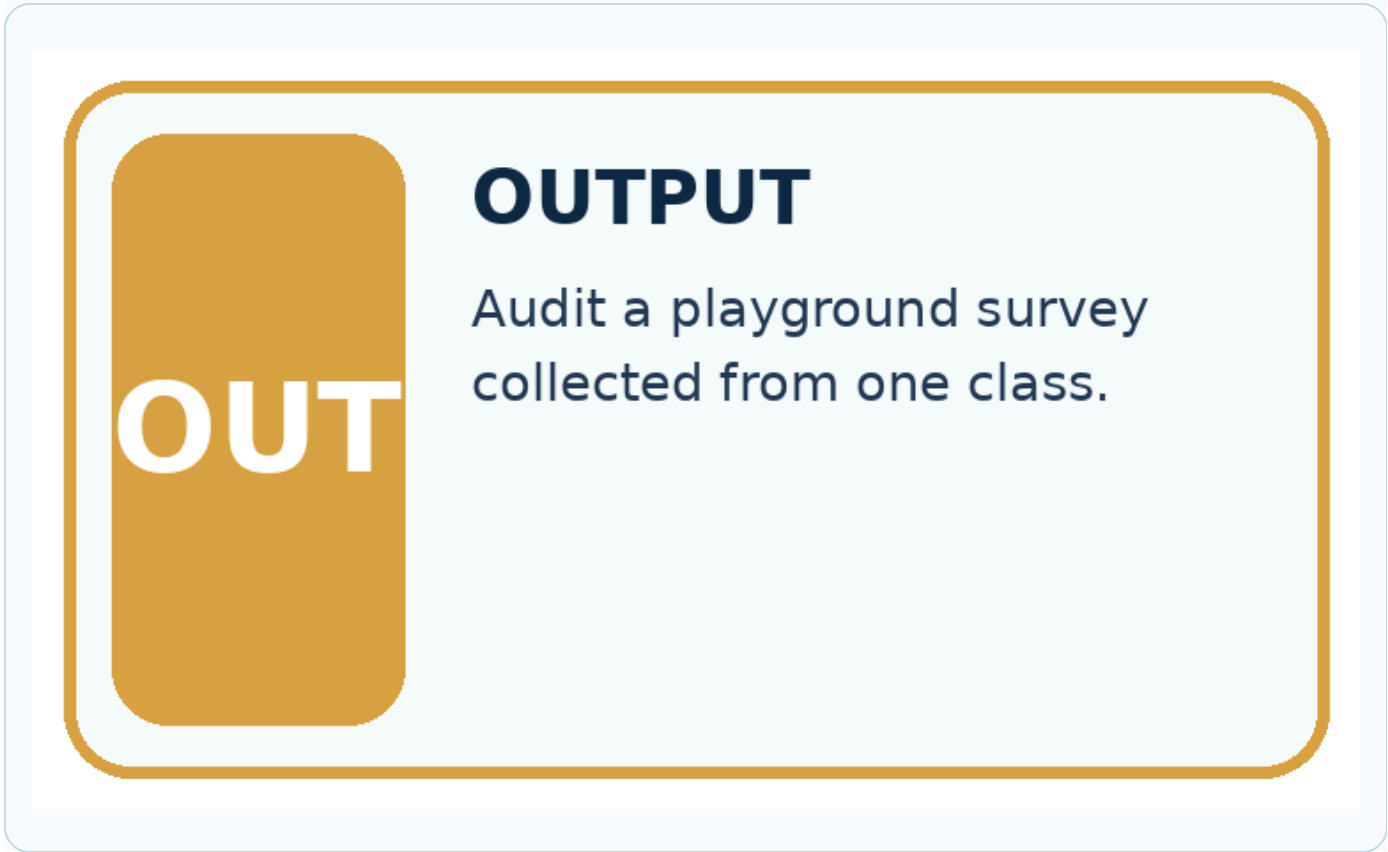
Trace the information. At each stage, ask what changed and how you could test it.



↓


PROCESS
A sample must reflect the people and situations where the system will be used.

↓


OUT
OUTPUT
Audit a playground survey collected from one class.

Which stage could fail even when the other two are correct? Explain:

> **Write or draw your reasoning and evidence.**

A closer explanation

Match the people to the question

Apply the connected idea: Graphs and honest scales

Make bar lengths mean something

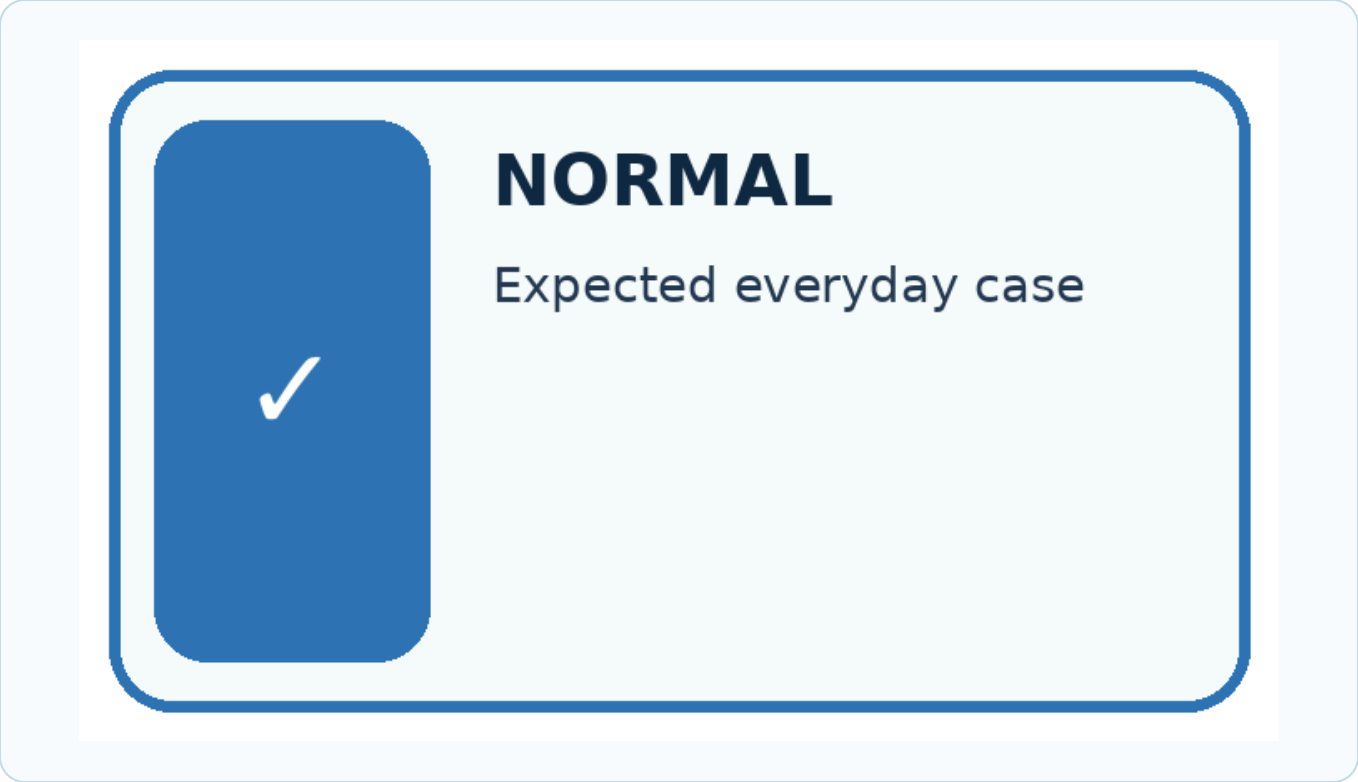
Connect the picture and the explanation

MAIN LESSON · SECTION 4

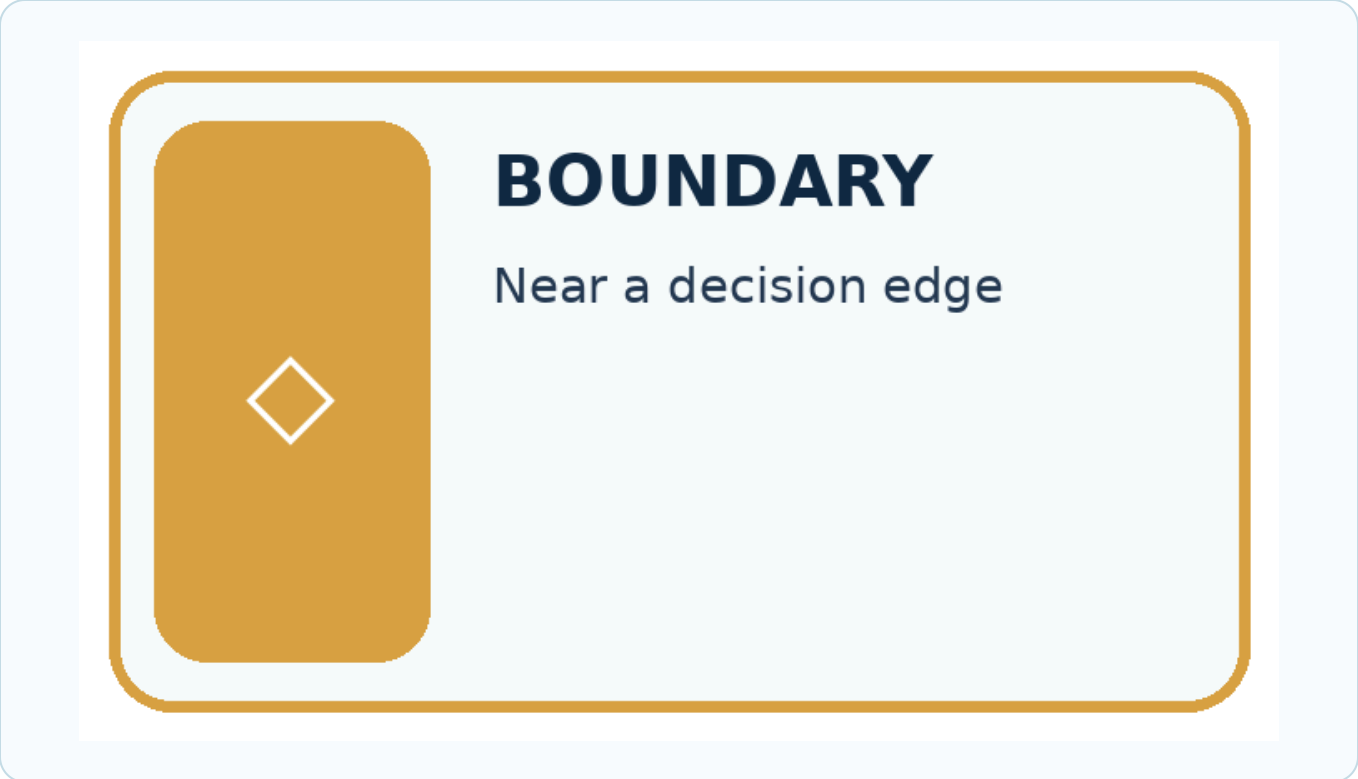
Compare collection plans and their coverage

> More replies from one club do not include the people outside it.

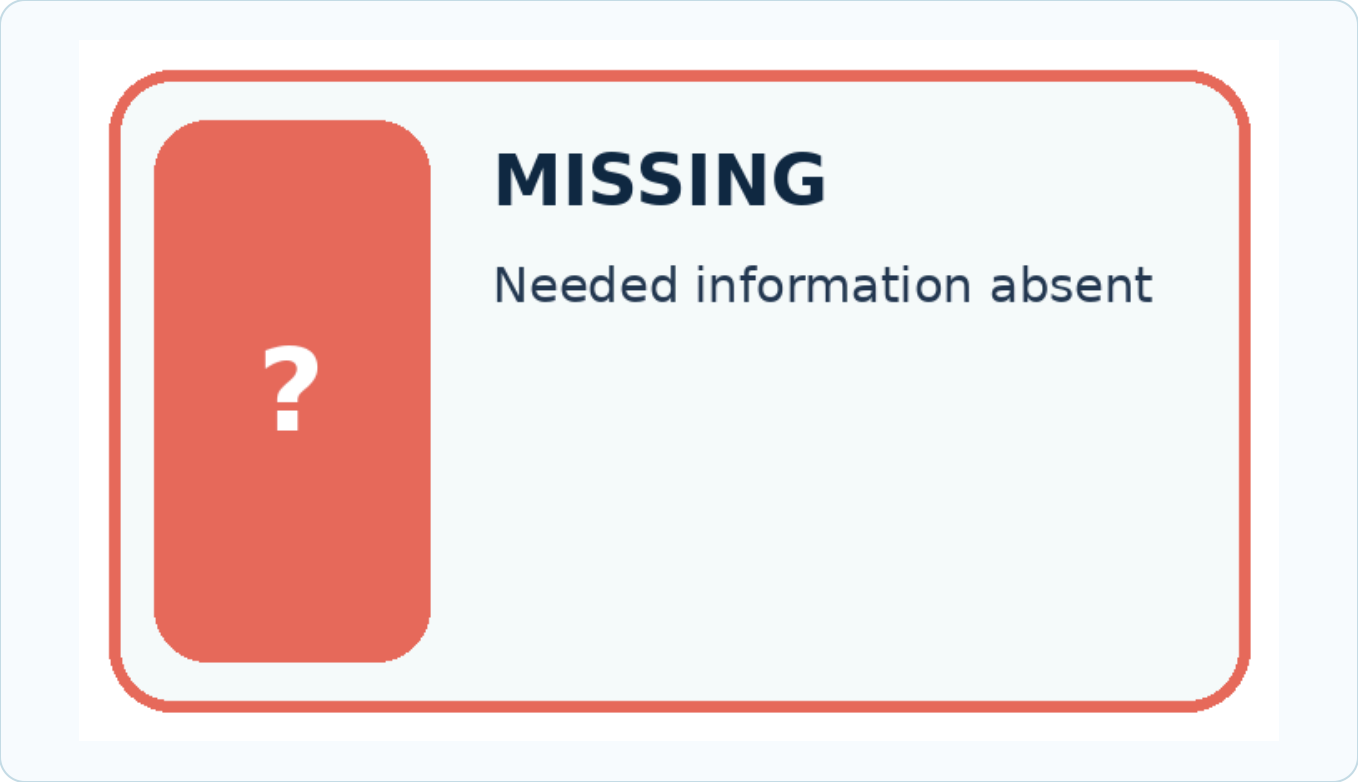
Four pictures with a purpose



Normal case: predict the result and deciding evidence.



Boundary case: identify the exact limit.



Missing information: name what must be clarified.



Transfer case: explain which procedure still applies.

Choose a revealing case

- Which case is most likely to reveal a hidden assumption? Why?

Write or draw your reasoning and evidence.

Predict the supplied changed case

- Change: Invite only the 15 library-club members instead.

▼ Check after predicting

Prediction: The invitation **process** no longer covers the whole grade through this frame.

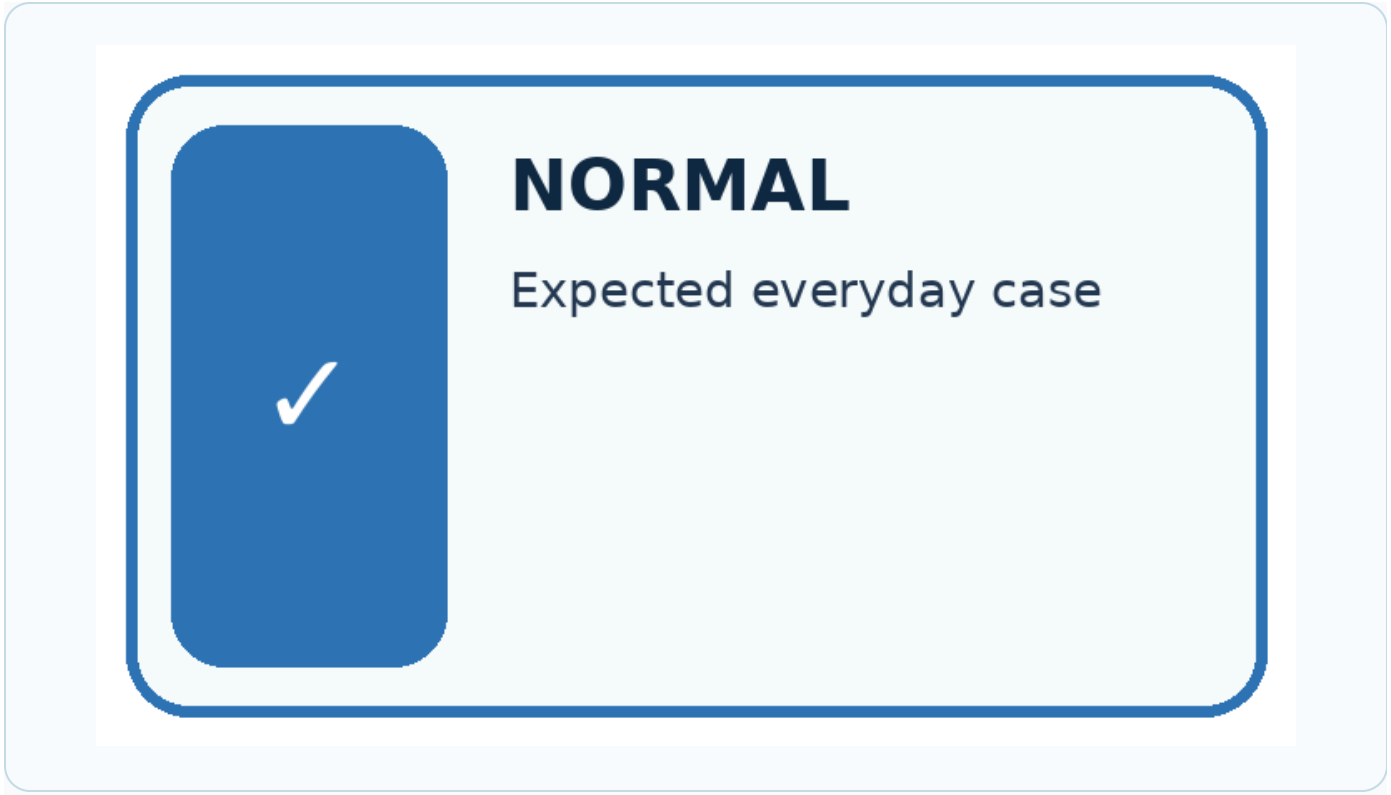
Reason: Club members may differ in reading interests; selection is now limited to one group.

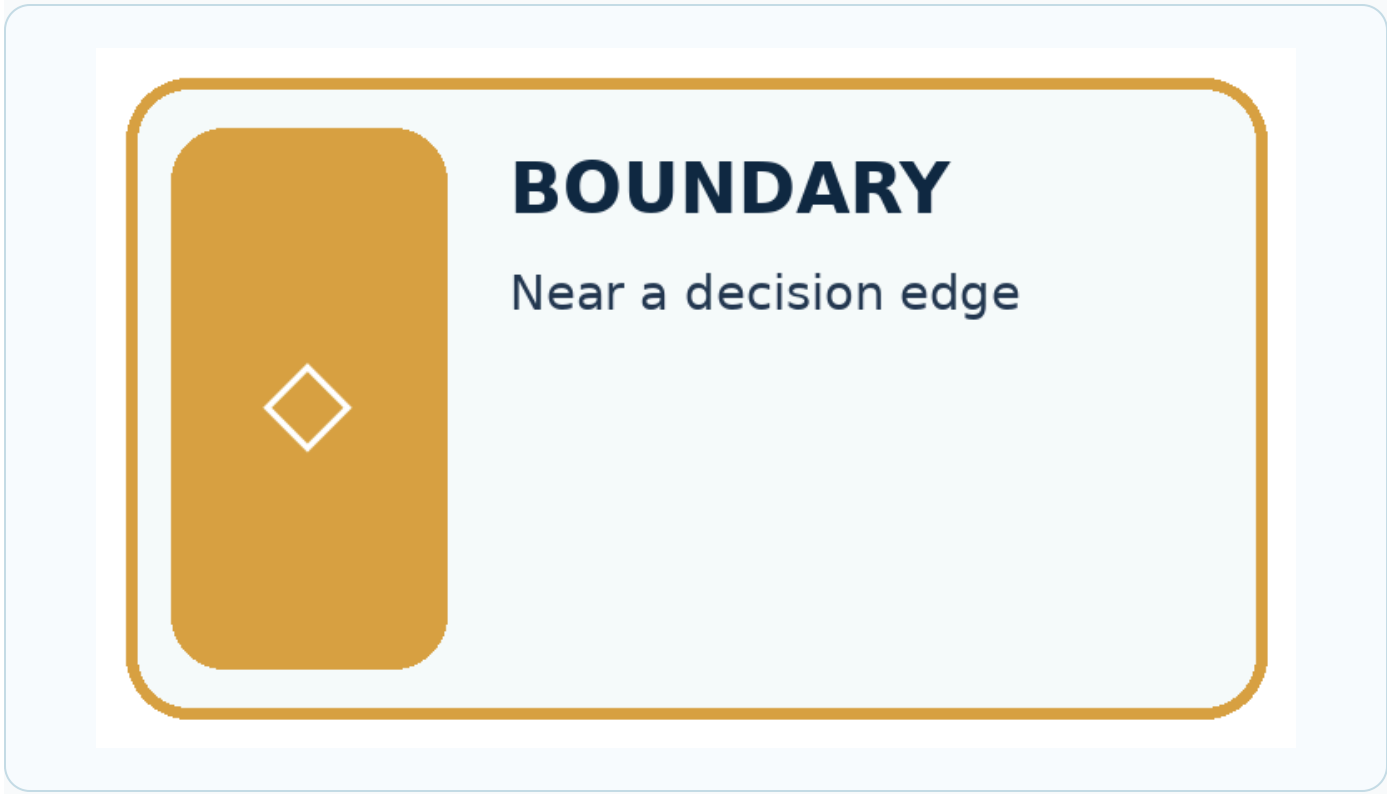
▼ Earlier wording, alternatives and picture transcripts

CASE GALLERY

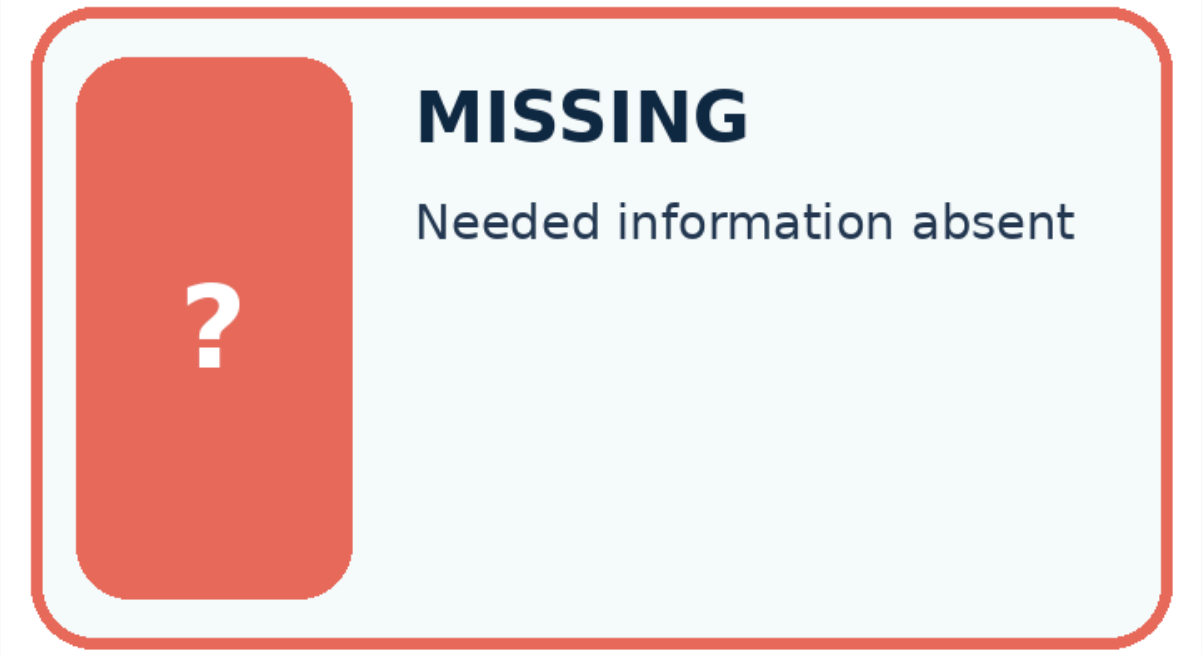
Four Tests, Four Kinds of **Evidence**

A system is not understood until it survives more than one comfortable example.





NORMAL CASE: predict and name **evidence**. BOUNDARY CASE: predict and name **evidence**.





MISSING INFORMATION: predict and name **evidence**. TRANSFER CASE: predict and name **evidence**.

Change one thing in the pictured case

MAIN LESSON · SECTION 6

Distinguish population coverage from invitation response rate

> Do not mix counts from different questions or respondent sets.

Question the extra-club-response shortcut

> **If the school adds 20 more club responses, which coverage problem remains?**

Keep the four examples separate

Separate example	What the supplied numbers describe
Coding-club activity poll	2 Code choices among3 club responses
W2 longer-lunch task	7 preferences among12 fictional respondents
Book-format invitation plan	Population60; invited15; responses12; no preference breakdown supplied
Graph extension activity poll	Drawing8, Building10;18 total responses

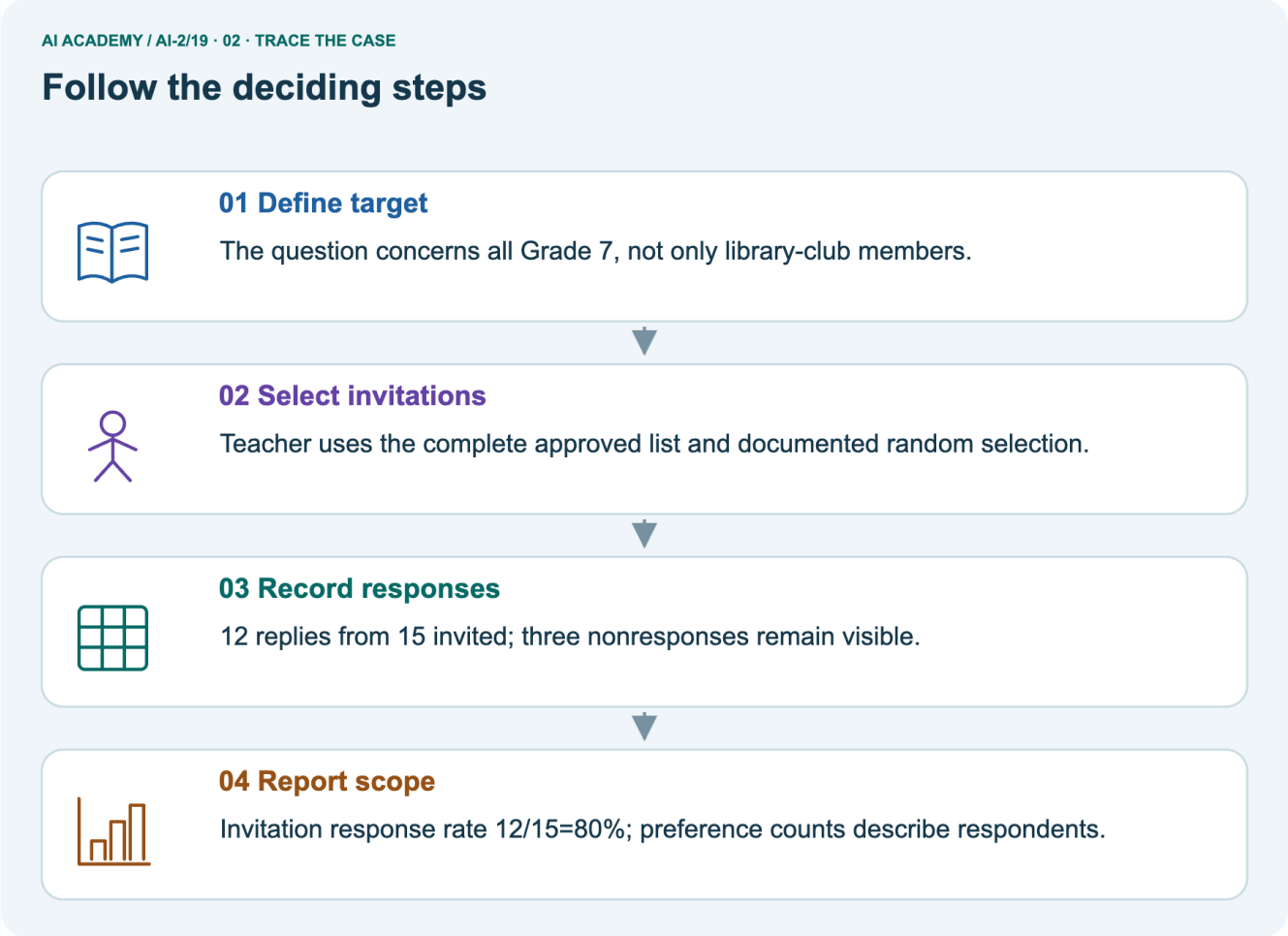
Read the separate book-format plan

A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

Population: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Trace frame, invitations and responses



Follow the labelled picture with the matching explanation. Open the vector diagram

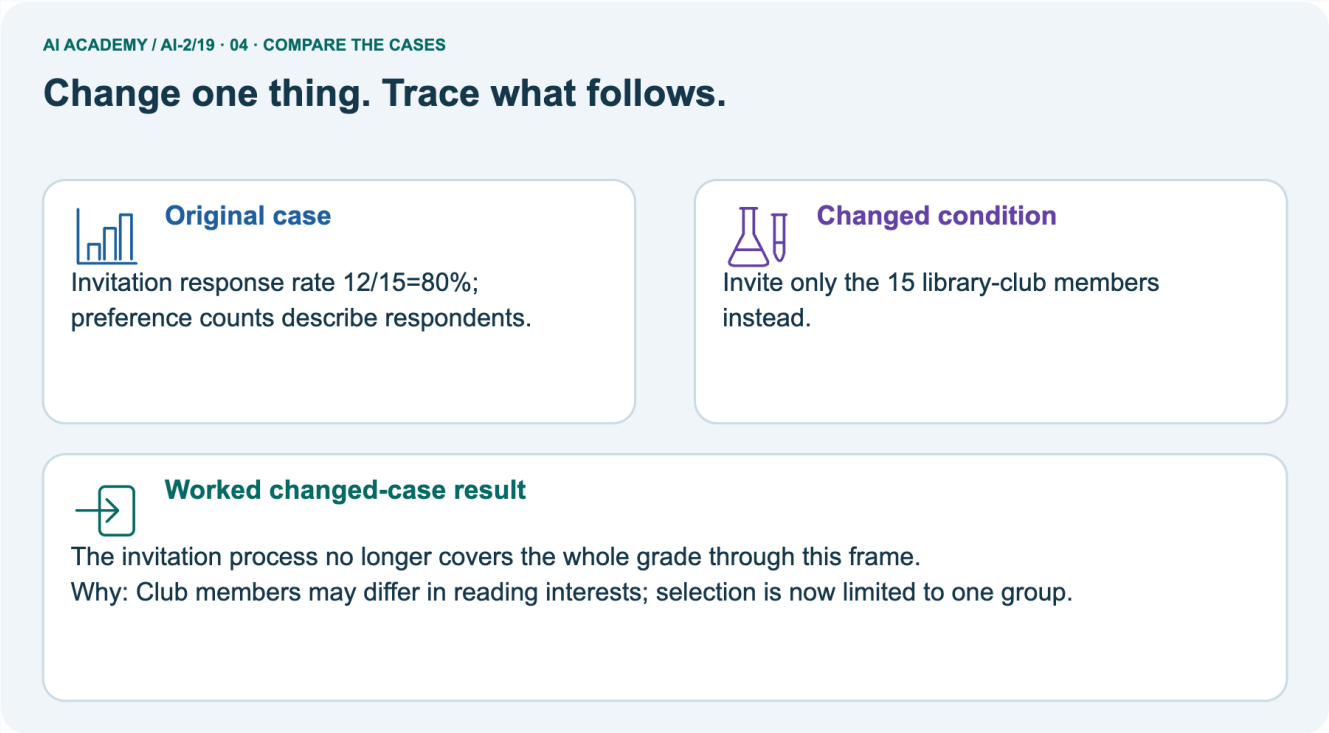
A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Predict the club-only invitation change

- Change one thing: Invite only the 15 library-club members instead. Predict the result before checking the explanation. What remains the same?

▼ Check after predicting



Follow the labelled picture with the matching explanation. Open the vector diagram

Explain the denominator and remaining limits

- Read the labelled picture: Can **80% response** be reported as **80%** of the whole grade responding?
- Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

▼ Optional graph extension · A separate4-versus6 comparison

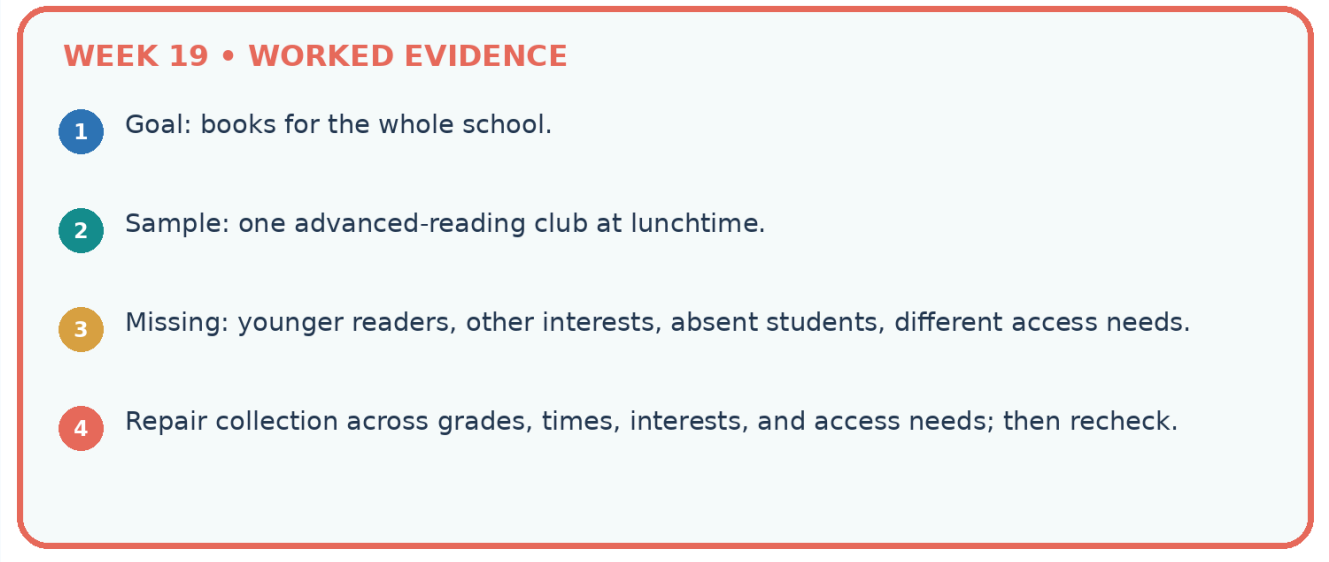
Draw bars for counts 4 and 6 using one square per vote. State the difference and ratio.

▼ Earlier wording, alternatives and picture transcripts

WORKED **EVIDENCE**

Reason Step by Step

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.



1. Goal: books for the whole school.

2. **Sample**: one advanced-reading club at lunchtime.

3. Missing: younger readers, other interests, absent students, different access needs.

4. Repair collection across grades, times, interests, and access needs; then recheck.

Recalculate, retrace, or restate the reasoning in your own **representation**:

> **Write or draw your reasoning and evidence.**

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

A closer explanation

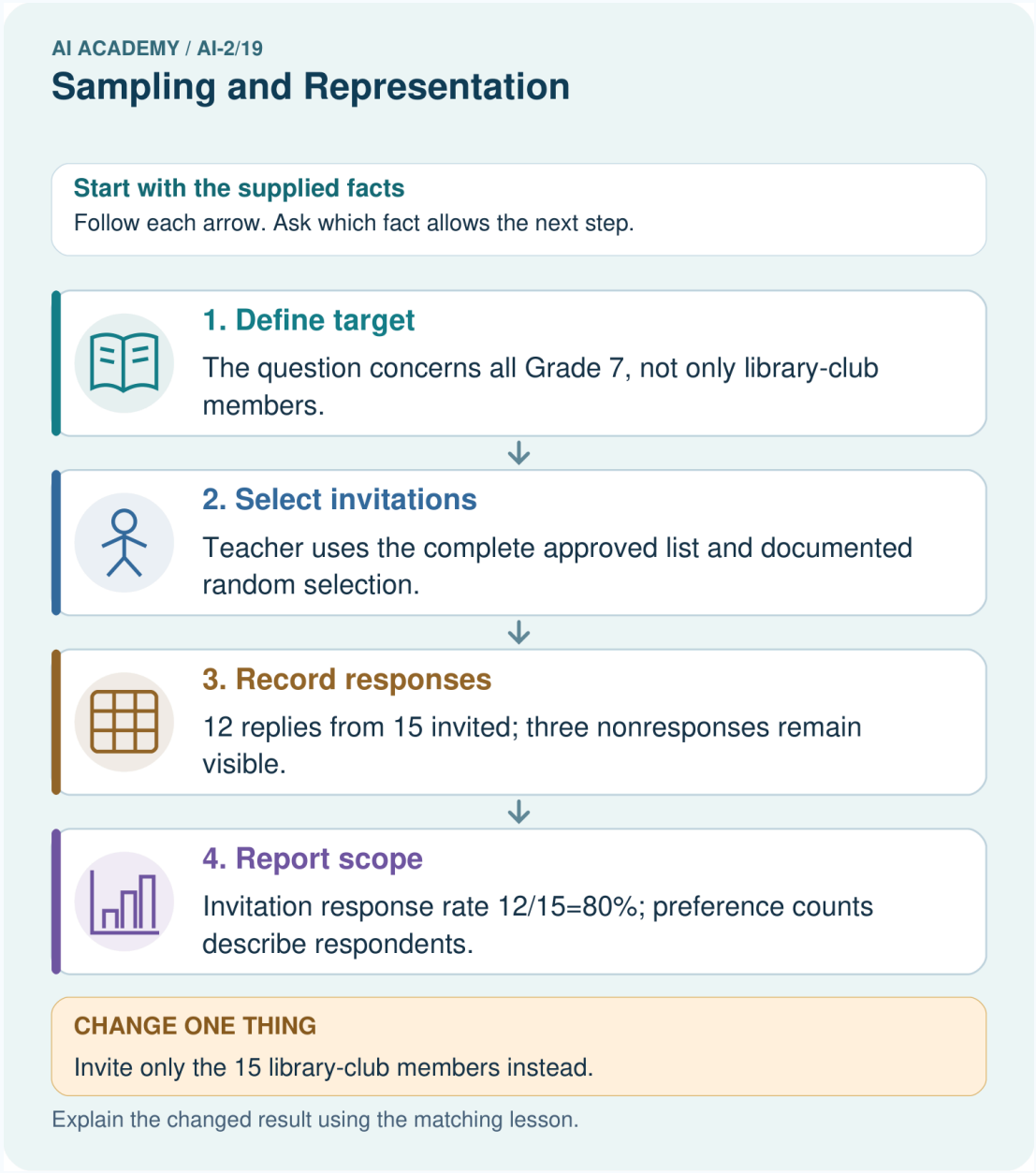
Apply the connected idea: Graphs and honest scales

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Sampling and Representation

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Define target	The question concerns all Grade 7, not only library-club members.
2. Select invitations	Teacher uses the complete approved list and documented random selection.
3. Record responses	12 replies from 15 invited; three nonresponses remain visible.
4. Report scope	Invitation response rate 12/15=80%; preference counts describe respondents.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

MAIN LESSON · SECTION 7

Fair invitations do not guarantee complete or unbiased response

> Report both invited and responding counts.

Explain selection and nonresponse separately

Selection and **response** are separate. Even a fair invitation **sample** can be distorted if only interested students respond. Report numbers invited and responding and explain who may be absent.

Answer the coverage question

> **Twenty students are selected fairly but only 8 answer. Does fair selection remove the need to report nonresponse?**

▼ Optional graph extension · Preference is not learning effectiveness

Counts alone do not show why students chose an activity or whether nonrespondents would choose differently. Graph choices should match the question: categories suit bars; ordered time **observation**s can suit a line.

> **Art has the most votes among 18 responses. Does that establish Art is best for learning?**

▼ Earlier wording, alternatives and picture transcripts

ERROR DETECTIVE

Compare → Diagnose → Repair

Do not patch the answer. Find the earliest unsupported assumption, wrong step, or weak **evidence**.

!

COMPARE • DIAGNOSE

Weak idea: A large sample from one narrow group represents everyone. Strong move: locate evidence, change one cause, and retest.

Evidence record	Expected	Actual	Likely cause / next test
normal case	works		
boundary case	stated rule		
missing input	safe response		
fresh transfer	generalizes		

Repair one cause. Why is this repair better than changing several things?

Write or draw your reasoning and evidence.

A closer explanation

Apply the connected idea: Graphs and honest scales

Narrow the claim or change the collection plan

> A new heading cannot repair a restricted sample.

Repair the claim while protecting identities

Changing the report heading from club **response**s to all students does not fix sampling. Change the collection plan or narrow the claim.

Avoid collecting names when anonymous counts are sufficient; a teacher can manage selection without publishing identities.

Practise the separate lunch result in W2

Use workbook W2 for the connected example “Questions **population**s and **sample**s”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge a successful-run guarantee

- Use the worked case for Questions **population**s and **sample**s. Proposed explanation: “A successful run proves the conclusion”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured response denominator

Can **80% response** be reported as **80%** of the whole grade responding?

This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ Optional graph extension · Zero baselines and labelled scales

A graph with bars beginning at 7 turns counts 8 and 10 into visible lengths 1 and 3. Redraw from zero and label counts directly.

A line chart can use a restricted **axis** for detail if clearly labelled, but it must not disguise its **scale**.

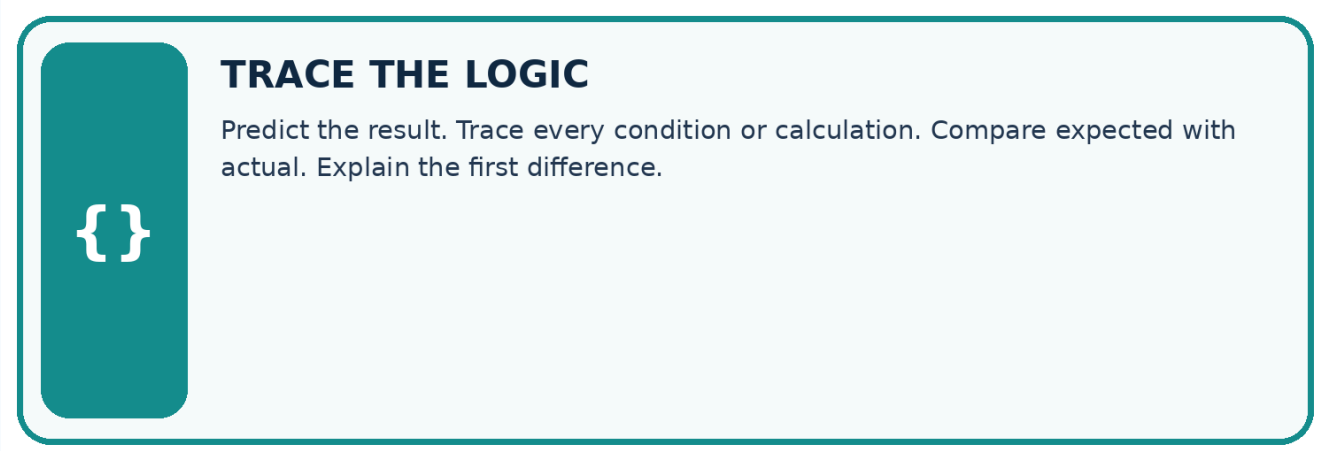
Use its named workbook W2. Identify the given **input**s, try the middle steps with less help, then explain your trace. Check this claim before group discussion: A successful run proves the conclusion

▼ Earlier wording, alternatives and picture transcripts

LOGIC STUDIO

Trace Before You Run

Block code, pseudocode, tables, and diagrams are different ways to expose the same reasoning.



SET **evidence**_record PREDICT expected_result RUN one normal case and one boundary case COMPARE actual_result with expected_result IF different: locate first cause; change one factor; RETEST EXPLAIN result, limitation, and next safe step

Trace one **input**. Which line changes the state or chooses a path?

Write or draw your reasoning and evidence.

A closer explanation

Apply the connected idea: Graphs and honest scales

Practise with less help: Read the sample result

Check your reasoning

Practise the connected topic: Graphs and honest scales

Explain the picture

My explanation using a pictured step and the given facts

MAIN LESSON · SECTION 9

Audit three complete fictional survey plans

Compare inclusion and nonresponse without inventing survey outcomes.

The three supplied plans

Plan to audit	Fictional facts supplied
A · Club volunteers	Target all60 Grade7; invites only15 coding-club members;12 reply

Plan to audit	Fictional facts supplied
B · Whole-frame invitations	Target all60 Grade7; complete fictional frame; randomly invite15;12 reply
C · All invited	Target all60 Grade7; invite all60 using the same neutral question;12 reply

Predict limits, calculate denominators and propose a repair

1. For each plan identify the target, who can receive an invitation, who answered and who remains unobserved. Predict its coverage limits before discussion. Record invitation response rates12/15,12/15,12/60 and distinguish each from the fraction of all60 who responded. PlanB improves invitation coverage but cannot guarantee representative respondents; PlanC still has nonresponse despite inviting everyone. No preference counts are supplied, so do not invent a winning activity. Propose one better documented collection or follow-up plan using fictional identifiers and neutral identical options. Keep the independent activity-survey design for a first attempt before feedback.

Keep the evidence and remaining question

Test / input · Prediction · Actual · Evidence / calculation · Change or keep?

> **Test / input:** normal
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** boundary
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____

> **Change or keep?:** _____

> **Test / input:** missing
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

> **Test / input:** fresh transfer
>
> **Prediction:** _____
>
> **Actual:** _____
>
> **Evidence / calculation:** _____
>
> **Change or keep?:** _____

What did the hardest test teach you that the normal test did not?

> **Write or draw your reasoning and evidence.**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ **Earlier wording, alternatives and picture transcripts**

EVIDENCE LAB

Predict → Test → Record → Explain

Keep expected and actual results separate. Surprises are useful data when recorded honestly.

PREDICT → TEST → RECORD → EXPLAIN

Compare sample cards with an intended-user checklist, highlight missing settings and users, and design a broader collection plan.

Audit the supplied three fictional survey plans. State population, invitation frame, selection method, invited and responding counts, response denominator and the remaining coverage/nonresponse limits. Do not conduct a real survey for this exercise.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: design the new activity-preference sample plan

> Name the target, frame, selection, neutral question and nonresponse report.

The independent design brief

Design an anonymous **sample** plan to ask all Grade 7 about three activity choices. State **population**, **sampling frame**, selection method, question, and how you will report non**response**.

Fictional-data and anonymity scope

Design the plan using invented identifiers and a hypothetical teacher-managed frame. You do not need to collect names, use a real school roster or run a survey. State how responses could be anonymous while invitation and response counts remain auditable.

Your first attempt and support record

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback after the attempt

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Plan independently

Week 19 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You survey only the chess club about all students' favourite books. Identify the sampling limitation and one improvement.

My answer, with a trace, calculation or supporting **evidence**

> **Write or draw your reasoning and evidence.**

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

> **Write or draw your reasoning and evidence.**

Connect to your project

Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

The **evidence** I will keep

> **Write or draw your reasoning and evidence.**

Transfer practice from the original programme

A survey about school reading preferences includes only volunteers from a mystery club. Explain why 100 **response**s may still fail to represent the school. Propose a better sampling plan using fictional data.

> **Explain the mechanism, evidence and one limit.**

Use feedback to improve

MAIN LESSON · SECTION 11

Document whose requests inform the recommender

> Coverage evidence is different from a representative prevalence estimate.

This week’s concrete portfolio contribution

Use invented reader scenarios to document who the recommender is intended to serve and which relevant settings are absent from the available examples. A scenario checklist can reveal gaps but does not estimate how common preferences are among real users. Do not collect personal attributes merely to fill the checklist. State whether evidence describes respondents, invited users or the target population, and keep accessibility-related needs optional and fictional in this lesson.

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users.

Keep the system and human choices visible

SYSTEM MAP Reader need → allowed **input**s → transparent rule or score → ranked candidates → explanation → test record → **human decision**

What will you add, change, or verify in the project this week?

Write or draw your reasoning and evidence.

What should remain under **human control**?

Write or draw your reasoning and evidence.

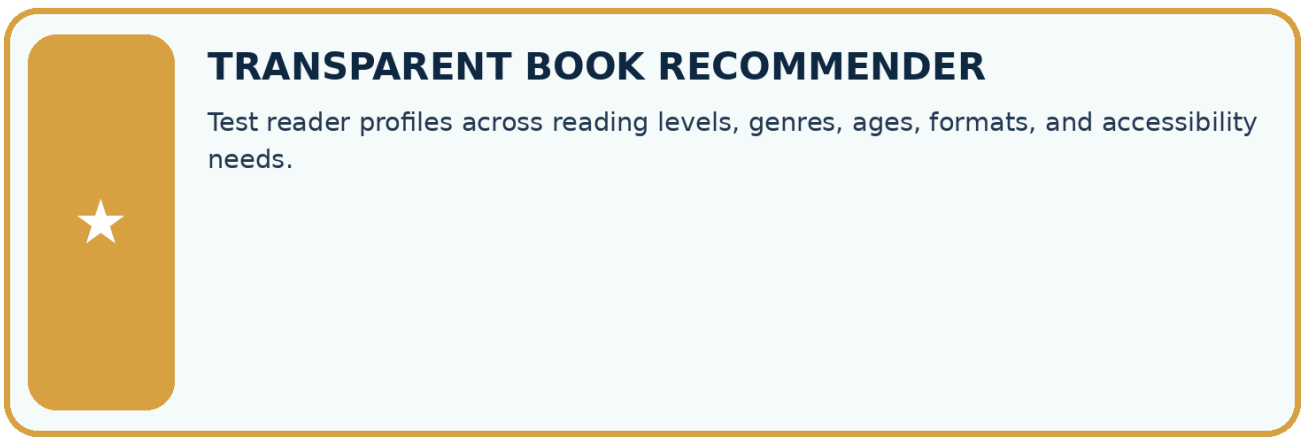
RESPONSIBLE DESIGN Collect only necessary information. Show uncertainty and limits.
Never confuse a ranking with a fact about a person.

▼ Earlier wording, alternatives and picture transcripts

CUMULATIVE PROJECT

Transparent Book Recommender

Every week adds one inspectable piece. The system must show how it reached a result.



THIS WEEK'S CHECKPOINT Test reader profiles across reading levels, genres, ages, formats, and accessibility needs.

Evidence for your project

MAIN LESSON · SECTION 16

Finish: state the population your evidence actually describes

> Keep the sample and response limits in the conclusion.

Explain without notes

- Close your notes. Explain this goal in your own words: Define a question and target **population**

My deciding step, evidence and one remaining question

Return on a changed plan

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Words for the main lesson and extension**

REFERENCE · SECTION 5

Reference · Words for the main lesson and extension

> Use each word with a concrete record or calculation.

Core words

Word	My own example	Not this / boundary
sample	A selected group of observations, or an outcome drawn from a distribution; the context identifies which meaning applies.	
population	The full group of cases or people a study aims to describe.	
representation	A way to encode information so a person or system can work with it.	
sampling bias	A systematic mismatch between selected examples and the population the study intends to describe.	

Main explanation terms

Word	Meaning
Sampling frame	The list or source from which a sample is selected
Response	A recorded answer from a selected participant

Extension terms

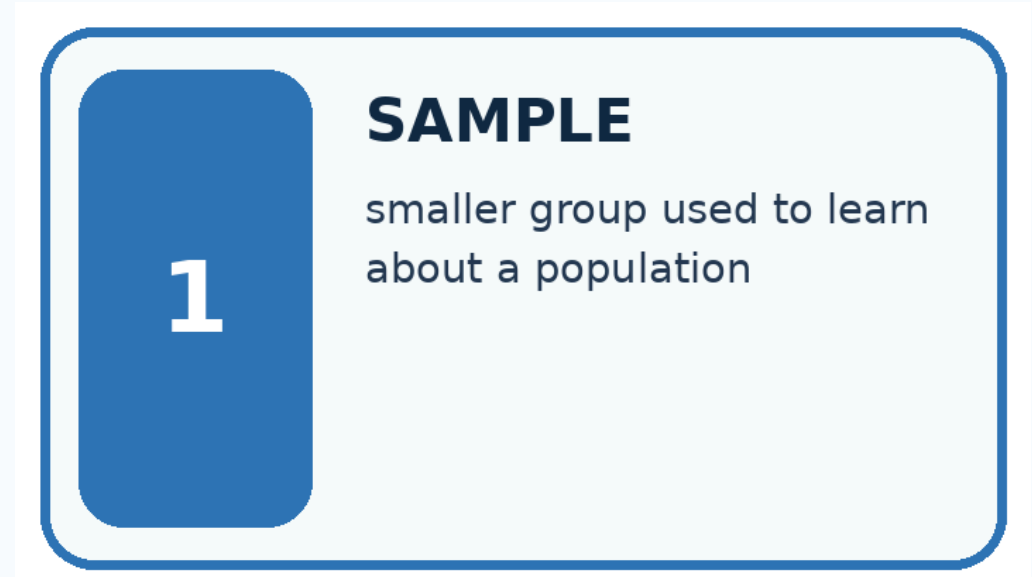
Term	Meaning
Axis	A reference line with a scale on a graph
Baseline	The starting reference value on a scale
Bar chart	A graph using bar lengths to compare categories
Scale	The mapping from values to positions or lengths

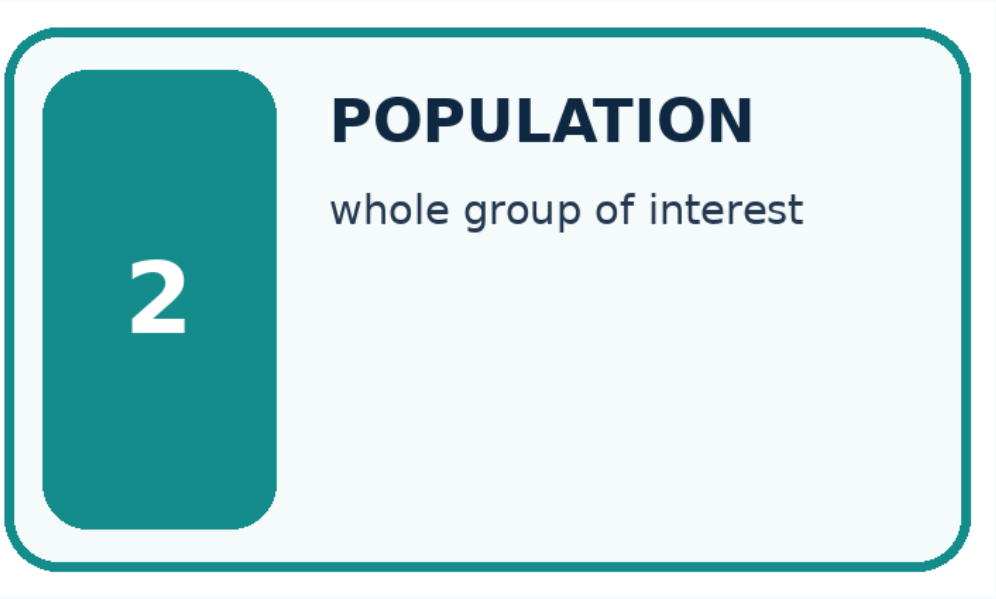
▼ Earlier wording, alternatives and picture transcripts

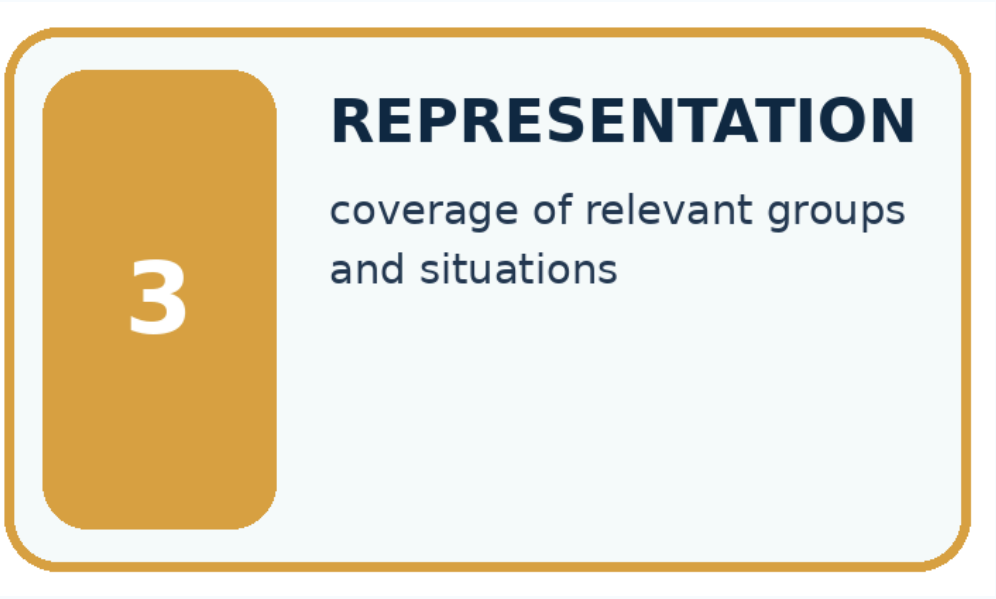
VOCABULARY LAB

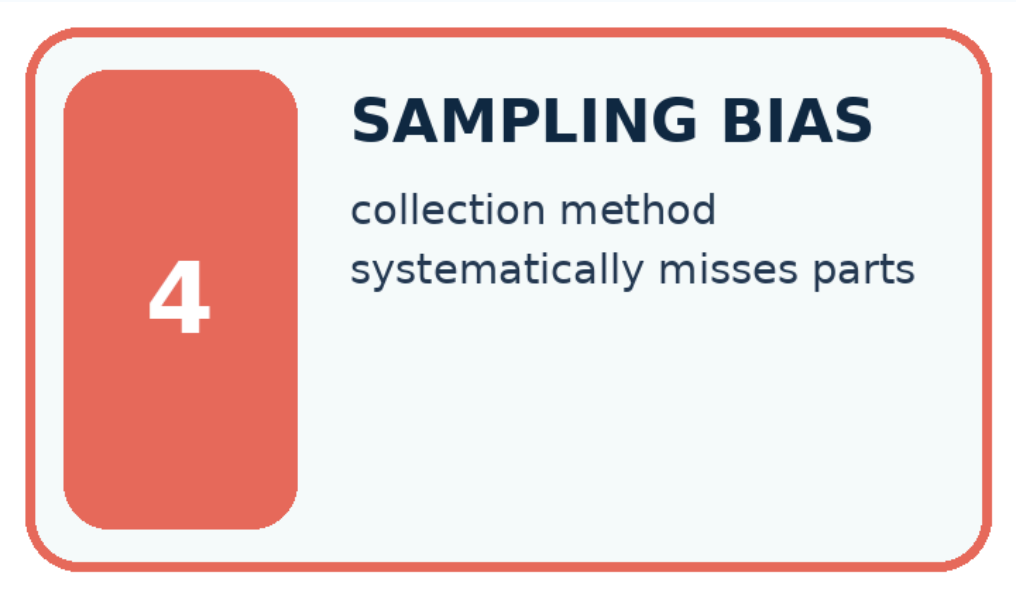
Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.




2
POPULATION
whole group of interest


3
REPRESENTATION
coverage of relevant groups and situations



Words used in the closer explanation

Terms for Graphs and honest scales

▼ **Extra challenge · Choose a stronger explanation**

REFERENCE · SECTION 10

Extra challenge · Choose a stronger explanation

> Choose the next useful challenge rather than repeating the ladder.

Thinking ladder

- 1 • NAME: Define the concept without repeating the textbook.
- 2 • TRACE: Show the information, state, branch, calculation, or **evidence** path.
- 3 • APPLY: Solve a different example independently.

- 4 • CHALLENGE: Find a boundary case, counterexample, or unfair outcome.
- 5 • EXPLAIN LIMITS: Say when the result should not be trusted or used alone.
- My strongest **evidence** of independent understanding:

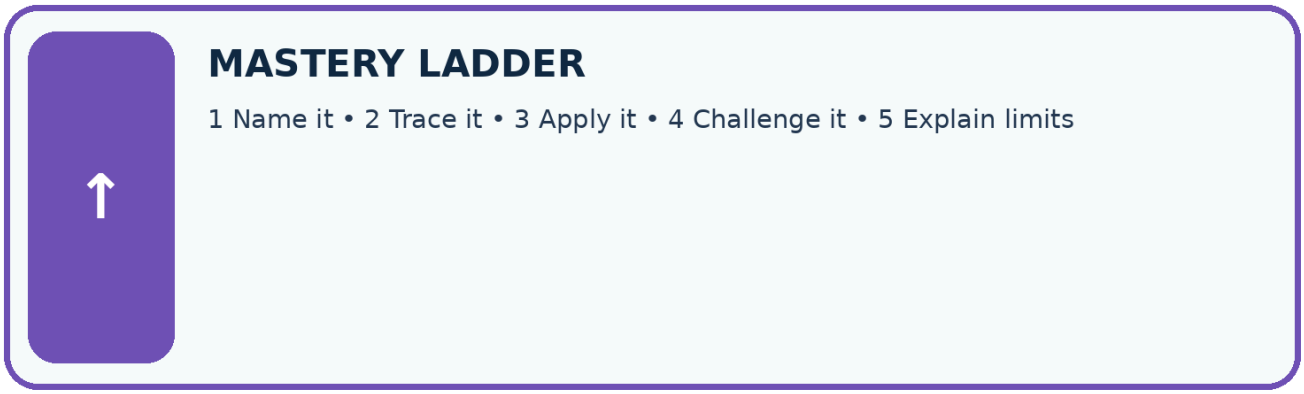
Write or draw your reasoning and evidence.

▼ Earlier wording, alternatives and picture transcripts

MASTERY

From Knowing the Word to Owning the Idea

Move upward only when you can produce **evidence** without copying the worked example.



▼ Reference · Check your explanation

Reference · Check your explanation

> Use after practice; Plan independently is the selected independent task.

Evidence and limits

- 1. My fresh example, trace, calculation, or design:
- 2. My **evidence** that it works (include a boundary or missing-data case):
- 3. One limitation, fairness concern, or situation needing human review:

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

> **Write or draw your reasoning and evidence.**

Self-check

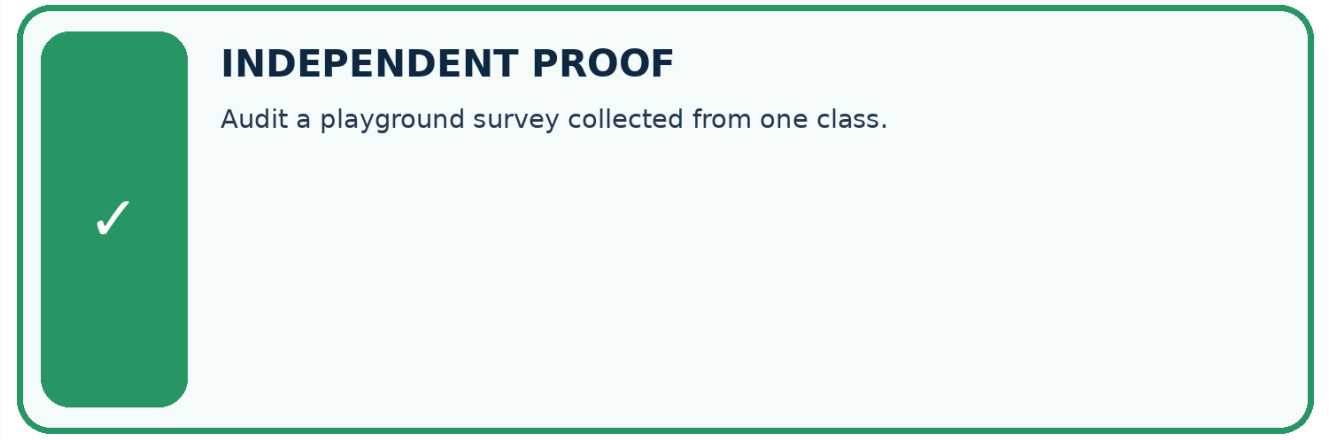
SELF-CHECK ☐ I explained the **mechanism** ☐ I used **evidence** ☐ I tested a fresh case ☐ I named a limitation ☐ I can answer 'why?'

▼ Earlier wording, alternatives and picture transcripts

INDEPENDENT MASTERY PROOF

Checklist for this week’s explanation

Use after practice. Plan independently is the selected independent task.



State the target, frame, invitation and response process, denominator and remaining limits.

▼ Extra practice · The earlier complete case

REFERENCE · SECTION 13

Extra practice · The earlier complete case

> Try the question before opening the worked explanation.

Question and given facts

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

Evidence or stage	Value or result
Sample	Adventure-club members
Intended population	School readers
Missing perspectives	Other interests and relevant grades

▼ Earlier wording, alternatives and picture transcripts

Week 19 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Who was left out of the survey?

Only adventure-club members answer a school reading survey.

Explain the sampling problem.

Membership selects for a relevant interest. Include relevant grades and interests with appropriate data practices.

▼ **Reference · Earlier checklist**

REFERENCE · SECTION 15

Reference · Earlier checklist

> Use when it helps inspect the evidence.

Checklist and first prediction

By the end, I can	How I will show it
Define a question and target population	Explain the deciding step, show a trace or test, and state one limit.
Describe how a sample was selected	Explain the deciding step, show a trace or test, and state one limit.
Limit conclusions to supported groups	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ **Earlier wording, alternatives and picture transcripts**

Start with this question: A fictional Grade 7 class wants to understand book-format preferences across its whole grade.

▼ **Extra practice · Pictures, feedback and repair**

REFERENCE · SECTION 17

Extra practice · Pictures, feedback and repair

> Try before opening feedback; keep the selected independent task separate.

First supported question

- A fictional Grade 7 class wants to understand book-format preferences across its whole grade.
- Which labelled step matches this detail? Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
- 1. Define target
- 2. Report scope
- 3. Record **response**s

- 4. Select invitations

- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ **Check after trying**

After your attempt, compare with the supported explanation: Report scope Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Changed-case question

- Changed-case practice: Invite only the 15 library-club members instead. Which conclusion is supported by the supplied facts?

- 1. This one example guarantees correct results for every future **input**.

- 2. The invitation **process** no longer covers the whole grade through this frame.

- 3. We can decide the result without examining the changed **input** or the procedure.

- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
>
> ---
>
> ---
>
> ---
>
> ---

▼ **Check changed case**

After your attempt, compare with the supported explanation: The invitation **process** no longer covers the whole grade through this frame. Club members may differ in reading interests; selection is now limited to one group.

Repair and keep evidence

Review Missing Data (ai-2/18). Goal: Detect absent and unusable values. Short practice: Why is the Boolean True rejected even though arithmetic can treat it like 1 in Python?. Return to this week and explain what you changed.

Review Over**fitting** Intuition (ai-2/16). Goal: Compare **training** and validation **evidence** for model capacity. Short practice: Does zero **training** error justify choosing Complex in the original record?.

Return to this week and explain what you changed.

Review Types of Classification Error (ai-2/13). Goal: Build a labelled confusion matrix from matched reference and **prediction** pairs. Short practice: Which error causes a sound book to be sent for unnecessary repair review?.

Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

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<summary>Reference · The six-picture journey</summary>

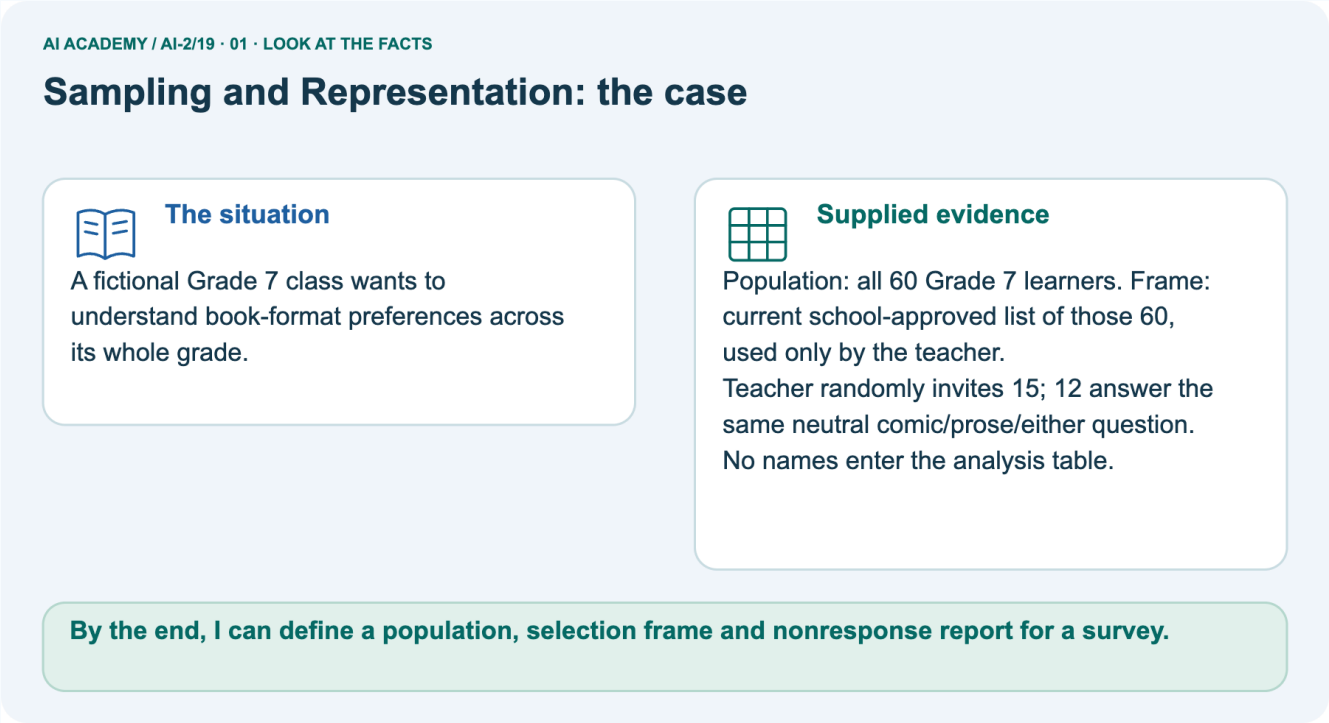
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REFERENCE · SECTION 18

Reference · The six-picture journey

> Revisit the picture that serves the current question.

01. Sampling and Representation: the case



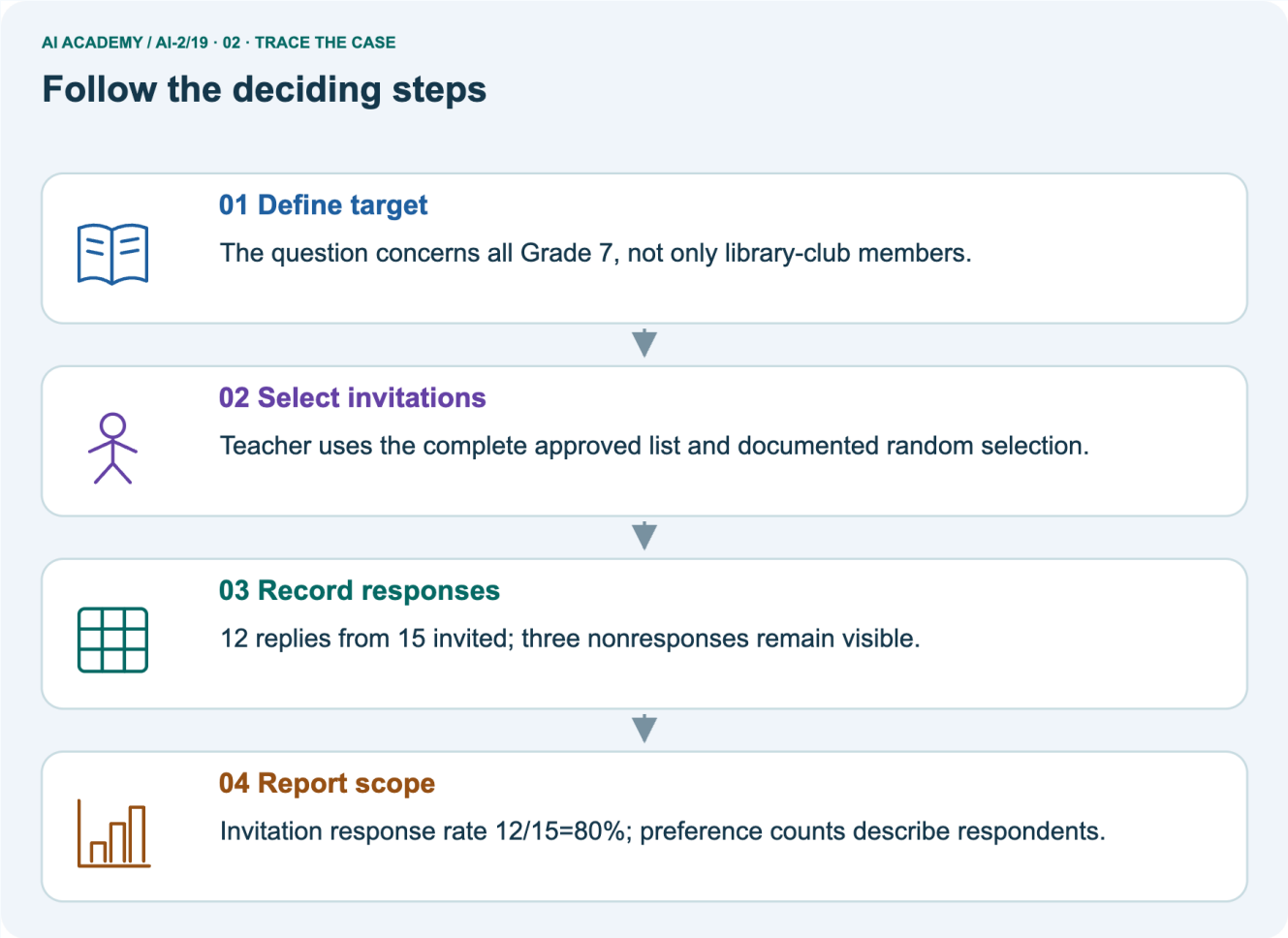
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional Grade 7 class wants to understand book-format preferences across its whole grade. Supplied facts: **Population**: all 60 Grade 7 learners. Frame: current school-approved list of those 60, used only by the teacher.

Teacher randomly invites 15; 12 answer the same neutral comic/prose/either question. No names enter the analysis table.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

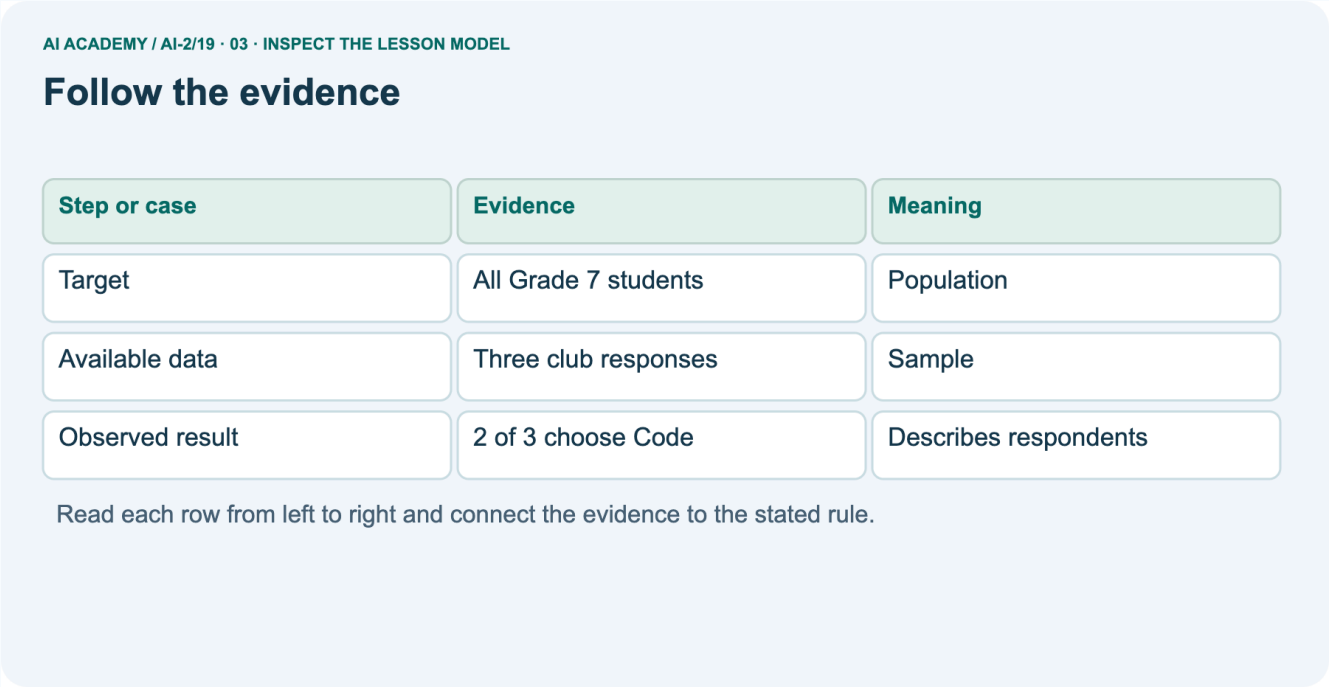


Follow the labelled picture with the matching explanation. Open the vector diagram

Define target: The question concerns all Grade 7, not only library-club members. → Select invitations: Teacher uses the complete approved list and documented random selection. → Record **response**s: 12 replies from 15 invited; three non**response**s remain visible. → Report scope: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.

Point to each numbered stage and explain which recorded **input** it uses.

03. Follow the evidence

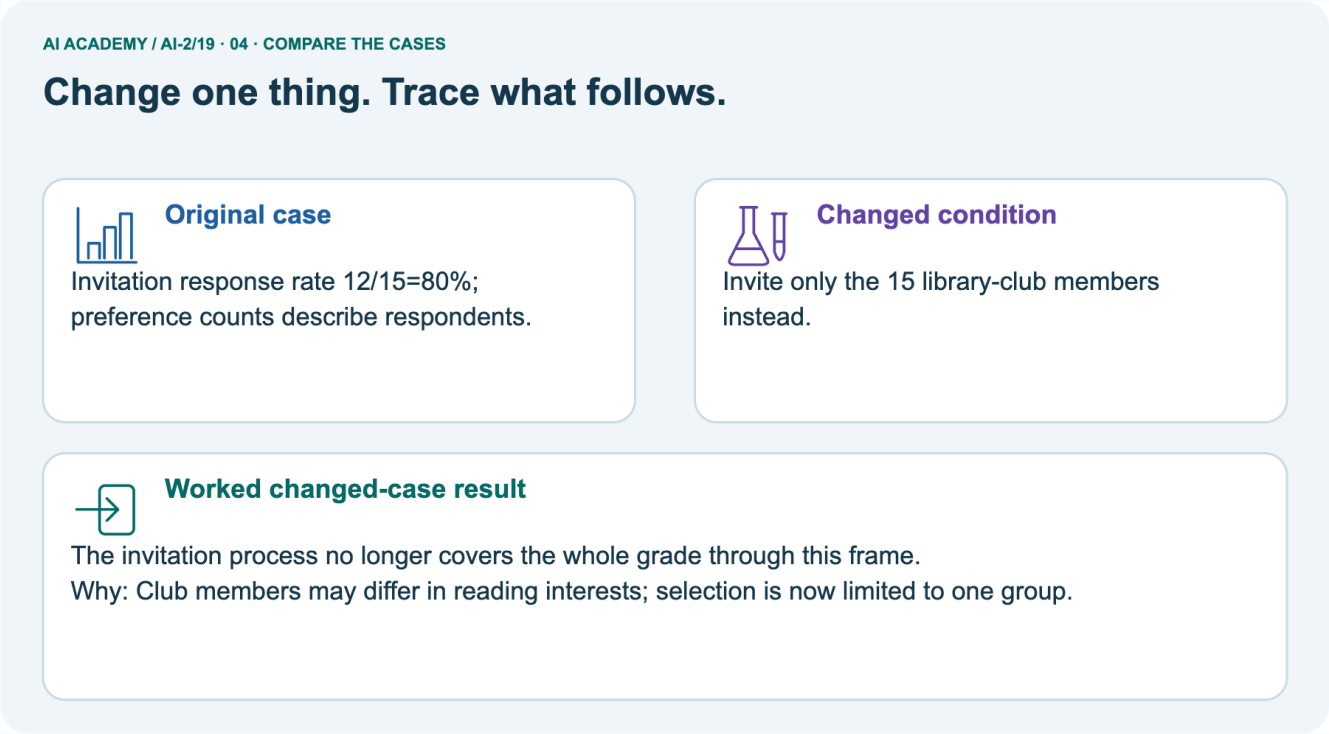


Follow the labelled picture with the matching explanation. Open the vector diagram

Follow the **evidence**. Target | All Grade 7 students | **Population**. Available data | Three club **response**s | **Sample**. Observed result | 2 of 3 choose Code | Describes respondents

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



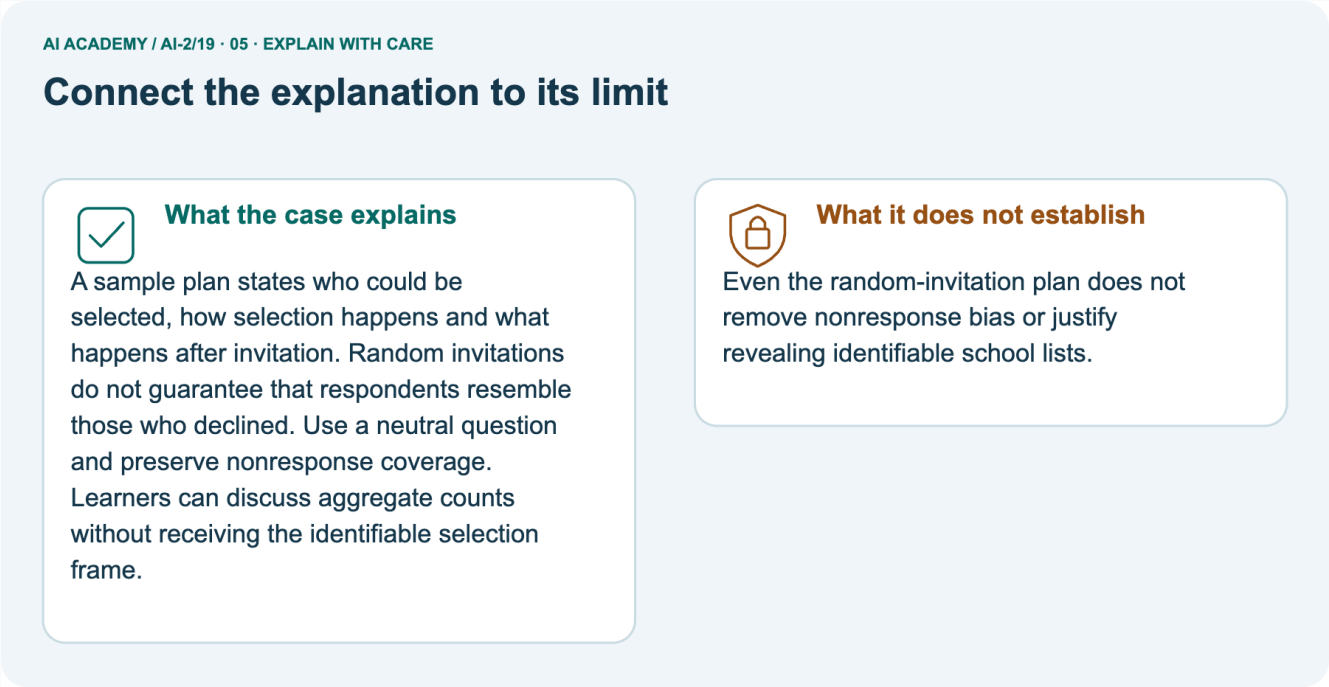
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Invitation **response** rate 12/15=**80%**; preference counts describe respondents.
Changed condition: Invite only the 15 library-club members instead.

Expected result: The invitation **process** no longer covers the whole grade through this frame.
Reason: Club members may differ in reading interests; selection is now limited to one group.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

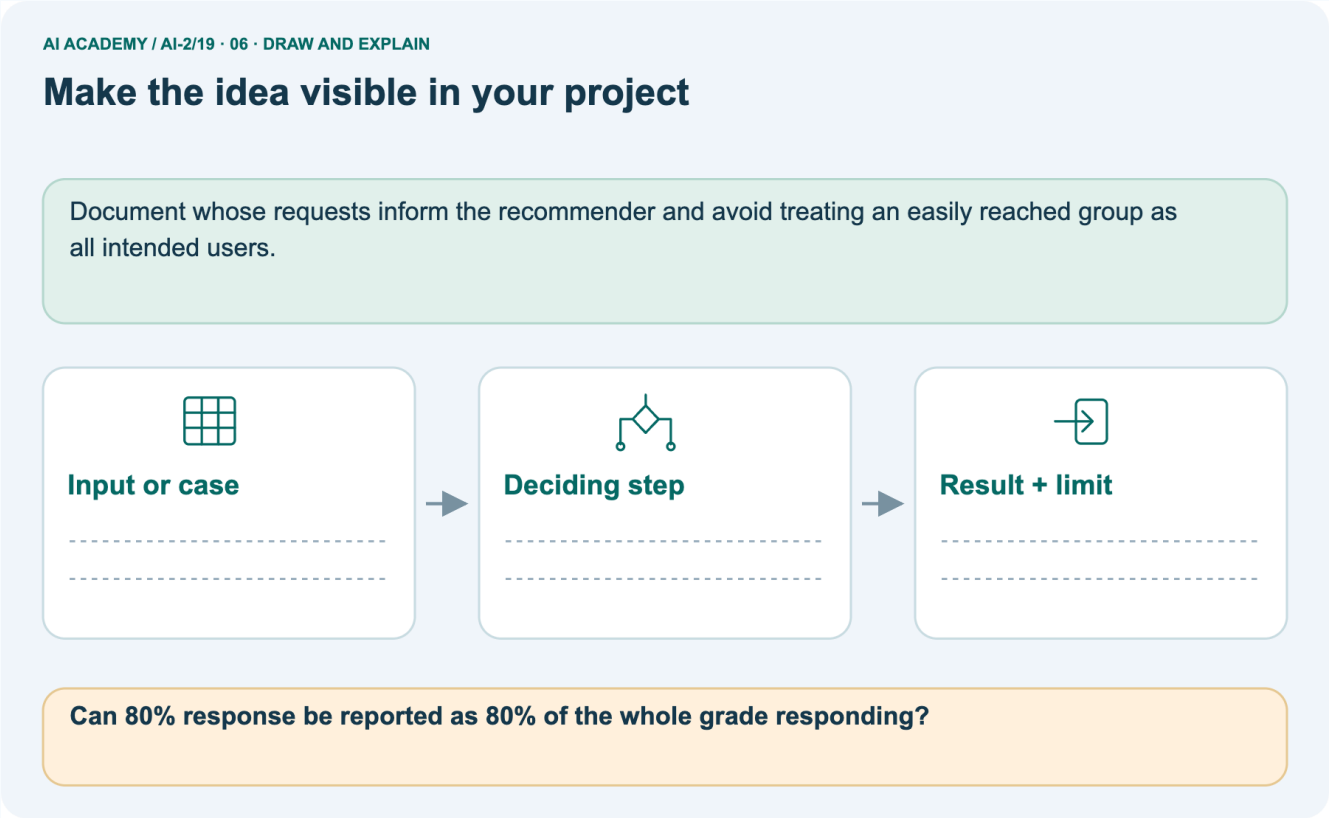
A **sample** plan states who could be selected, how selection happens and what happens after invitation. Random invitations do not guarantee that respondents resemble those who declined.

Use a neutral question and preserve non**response** coverage. Learners can discuss aggregate counts without receiving the identifiable selection frame.

Limit: Even the random-invitation plan does not remove non**response** bias or justify revealing identifiable school lists.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Document whose requests inform the recommender and avoid treating an easily reached group as all intended users. Draw your **input**, deciding step, result and limit.

Supported question: Can **80% response** be reported as **80%** of the whole grade responding?

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Sampling and Representation: the case

02. Follow the deciding steps

03. Follow the evidence

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.